

Ranak Mandal

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SKILLS SUMMARY

- **Languages:** Python
- **Frameworks:** Pandas, NumPy, Matplotlib
- **Tools/Platforms:** Proteus, AutoCAD
- **Soft Skills:** Communication, Critical Thinking

PROJECTS

- **Arduino – IOT Smart Home** October 2024
Designed and implemented a smart home environment enabling wireless control of electrical appliances such as lights and fans. Utilized an Arduino Uno microcontroller integrated with a Bluetooth module (HC-05) to receive commands from a smartphone. The system provided basic automation functionality, enhancing convenience and energy efficiency. Developed the hardware connections, embedded code in C/C++ via Arduino IDE, and tested the prototype to ensure responsive and reliable control.
- **ESP32 – Contactless Thermometer:** April 2025
Developed a non-contact thermometer capable of accurately measuring body temperature using the MLX90614 IR sensor and ESP32 microcontroller. Leveraged the ESP32's Wi-Fi capabilities for potential IoT integration and future remote data logging or alert systems. Implemented sensor interfacing and data processing using Arduino IDE, ensuring reliable temperature readings suitable for health monitoring applications.
- **Natural Paint Monitoring System** April 2025
Contributed to a multidisciplinary project focused on producing natural paint by designing and implementing the IoT hardware system. Integrated various sensors—including pH sensor, ultrasonic sensor, temperature & humidity sensor (DHT11/DHT22), and load cell—to monitor critical parameters during the paint-making process. Used microcontroller-based interfacing (e.g., Arduino or ESP32) to collect real-time data, enabling better tracking and analysis. Collaborated with a chemistry team responsible for the formulation while focusing solely on the hardware and sensor integration.
- **RC Vehicle with Assistive Sign Language Translation** March 2025
Designed and developed a remote-controlled (RC) vehicle equipped with robotic "helping hands" powered by servo motors, aimed at assistive communication. Integrated a camera module to capture hand gestures, which were then processed (via external software or systems) to translate sign language into spoken audio, enhancing communication for hearing- or speech-impaired individuals. Focused on hardware implementation using Arduino for motor control and gesture mimicry.
- **Stock Market Analysis & Prediction Tool** March 2025
Created a desktop application for stock analysis and prediction using Python. Integrated the Yahoo Finance API (yfinance) to fetch historical stock data and visualized it with Matplotlib, including candlestick charts using mplfinance. Built an intuitive GUI with Tkinter for user interaction and implemented a predictive model using RandomForestRegressor from scikit-learn to forecast stock prices. Evaluated performance with multiple error metrics (MAPE, MAE, RMSE, R²), allowing users to analyze trends and get basic price

CERTIFICATIONS

- VLSI for Beginners by NIELIT Calicut January 2025

INTERNSHIP

- Goldman Sachs (via Forage) – Micro-Internship Simulation February 2025

EDUCATION

- **Lovely Professional University** Punjab, India
Bachelor of Technology - Electronics and Communication Engineering Since August 2024
- **Oriental Public School** Nadia, West Bengal
Intermediate; Percentage: 78% April 2022 - March 2023
- **Oriental Public School** Nadia, West Bengal
Matriculation; Percentage: 93% April 2020 - March 2022